# JAY M. SEN

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## Education

**Rensselaer Polytechnic Institute (RPI) | Troy, NY** Expected May 2028

Bachelor of Science, Mechanical Engineering, *3.5 GPA*

*Rensselaer Medalist*

Relevant Coursework: Solid Mechanics, Numerical Programming, Thermodynamics, Calculus I-III, Differential Equations, Linear Algebra, Statics, Material Science, Physics I-II, CAD, Engineering Processes, Applied Statistics, Chemistry I-II

## Work Experience

**Production Engineering Intern | Cleaveland/Price Inc. | Trafford, PA** May 2026 - August 2026

- Conducted 180+ time studies of machining operations on the manufacturing floor, capturing cycle times, setup times, and process variables to support facility planning, inform equipment layout decisions, and optimize part workflow using lean manufacturing principles
- Developed 10+ templates and 2 AI agents to streamline and standardize the time study process, improving data consistency and repeatability across operators, machines, and parts to better translate findings into process and layout recommendations for production and engineering teams
- Built 2 calculators to estimate job times based on part characteristics and process parameters, enabling more reliable quoting and scheduling
- Audited job times to identify discrepancies between standard and actual times, discovering opportunities for process improvement and part cost savings

**Rensselaer Motorsport (FSAE) | Troy, NY** August 2025 - Present

- Currently part of the aero team leading a group working to optimize multi-element wing design
- Conducted Finite Element Analysis on rear wing truss using SolidWorks Simulation, improving Factor of Safety by 1.4
- Manufactured jigs, suspension components, and aero package components
- Served as race marshal to ensure the safety of drivers and team members by monitoring race conditions and responding to debris on the track

**Director of IT | Sen Engineering Solutions, Inc. | Monroeville, PA** June 2024 - Present

- Built and launched a fully dynamic Next.js website to repair site functionality and modernize brand identity
- Diagnosed and resolved hardware and software issues regarding computers, security systems, and online meetings
- Participated in contract and planning discussions with manufacturers and customers

**Dining Room Server | Juniper Communities | Monroeville, PA** February 2024 - August 2025

- Delivered meals to 60+ residents promptly and accurately according to dietary restrictions and resident preferences
- Fostered a compassionate, positive, and respectful environment for residents during meal times
- Maintained a clean and organized dining room following health and safety protocols
- Communicated regularly with nursing staff to address resident needs or concerns
- Conducted table setup, food prep, and kitchen support

## Skills

Python, Siemens NX, SolidWorks, SolidWorks Simulation, Geometric Dimensioning &Tolerancing, Mastercam, Manual Machining, Welding, Lean Manufacturing, Product Development and Launch, Customer Service, Google & Microsoft Suite

## Extracurricular Activities

**Eagle Scout | Scouting America** September 2013 - January 2026

- Eagle Scout Service Project: planned and built a 10 x 3 ft trail bridge in Boyce Park, Monroeville, PA
- Leadership: Senior Patrol Leader (Feb 2023 - Jun 2025), Patrol Leader (Feb 2022 - Feb 2023)
- Community Service (50+ hours): Community cleanups, trail maintenance, and summer camp maintenance

**Cello Performance** January 2011 - Present

- 14+ years of private lessons, group lessons, and performances with Rana's Music Studio
- Rensselaer Orchestra
- Pennsylvania Music Educators Association Region 1 finalist (2019)
- Gateway High School Orchestra Performance at Disney Springs, Lake Buena Vista, FL - Principal Cellist (2022)
- Pittsburgh Symphony Orchestra Side-by-Side alternate cellist (2024)